

RESEARCH PROFILE

My research combines experimental immunology, binding assays, and computational analysis of cellular responses. I ran **Biacore SPR and ELISA** assays for therapeutic antibody selection and coordinated external preclinical studies. I also develop single–cell and multimodal analysis workflows.

POSITION–RELEVANT EXPERTISE

- Binding measurements.** I ran Biacore SPR for association and dissociation kinetics, affinity ranking, receptor blockade, and human–primate cross–reactivity. I used ELISA binding, competition, and epitope assays alongside these measurements.
- Antibody candidate selection.** I compared antibody candidates in a humanized anti–CD40 program and followed lead selection with non–human–primate studies. The program supported a registered antibody patent in 2024.
- CRO coordination.** I designed and placed external studies, specified assay conditions, reviewed raw data, and tracked delivery dates. I used the results to make decisions about advancing candidates.
- Cellular and computational analysis.** I used high–parameter flow cytometry, single–cell and spatial profiling, and targeted proteomics to examine cellular responses. I wrote analysis code and tested results in held–out data.
- Grant writing and project execution.** I participated in proposal writing and project execution for industry projects totaling about \$1.3 million. I would consider LSRF and K99/R00 applications after confirming eligibility and project restrictions.

EDUCATION

DEGREES

Ph.D. 2014–2023	Seoul National University, Seoul, Republic of Korea Department of Biomedical Sciences <i>Thesis: Exploration of the Role of Regulatory T cells in Various Pathogenic Contexts</i> Advisor: Chung–Gyu Park, M.D., Ph.D.
M.S. 2012–2014	Seoul National University, Seoul, Republic of Korea Department of Biomedical Sciences <i>Thesis: Enhanced LPS–Induced Monocyte Activation by Recombinant Human Soluble CD14</i> Advisor: Chung–Gyu Park, M.D., Ph.D.
B.S. 2008–2012	Sungkyunkwan University, Suwon, Republic of Korea Department of Genetic Engineering

RESEARCH & PROFESSIONAL EXPERIENCE

Researcher (postdoctoral) 2025–present	Yonsei University Wonju College of Medicine · Dept. of Physiology / Organelle Medicine Research Center, Wonju, Republic of Korea Developed deep–learning approaches that combine imaging, transcriptomic, and clinical data in predictive models.
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**Associate Director
and Manager**
2021–2025

PB Immune Therapeutics Inc. · Dept. of Basic Research, Seoul, Republic of Korea

Associate Director, 2023–2025; Manager, 2021–2023.

Led a humanized anti-CD40 antibody program from candidate comparison through non-human-primate studies. Ran Biacore SPR and ELISA binding, competition, and epitope assays. Compared receptor blockade and human-primate cross-reactivity.

Designed and placed CRO and core-facility studies, reviewed data, and made candidate-advancement decisions. Managed milestones and regulatory documentation. Ran cynomolgus and marmoset EAE studies for multiple sclerosis programs.

Participated in outsourced antibody and AAV viral-vector manufacturing collaborations involving Samsung Biologics and WuXi. Took part in early joint-development research with Merck, CSL, and Johnson & Johnson.

Built in-house single-cell, spatial, and proteomic analysis workflows. Wrote KDDF proposals and carried out the funded lead-selection work.

Research Student
2020–2023

Single Cell Atlas of Immune Diseases (SCAID), Genomic Medicine Institute, Seoul National University, Seoul, Republic of Korea

Contributed to the founding proposal and work within the funded program. Took part in milestone tracking, multi-site sample quality control, and coordination across principal investigators. The program covered more than 700 donors, eight organs, and 23 immune diseases.

Researcher
2019–2021

Xenotransplantation Research Center, Seoul National University, Seoul, Republic of Korea

Processed rhesus macaque tissues and performed longitudinal immune monitoring in transplantation studies.

Teaching Assistant
2014–2016

Seoul National University College of Medicine · Dept. of Microbiology and Immunology, Seoul, Republic of Korea

■ SELECTED RESEARCH OUTPUTS

- 1 **Antagonistic Anti-CD40 Humanized Antibody.** *Republic of Korea patent*, Appl. No. 10–2024–0085972, registered 2024–07–01.

I ran SPR and ELISA measurements for candidate selection, receptor-blocking confirmation, and human-primate cross-reactivity. I coordinated external studies and led the antibody program through non-human-primate testing.

- 2 Woogil Song, **Seong-Jun Kang**, Taehee Stella Kim, Seyoung Jung, Sung Hee Kim, Su Chan Park, Sung Hae Chang, Eun Young Lee, Tae-Gyun Kim, Hyun Je Kim, Jeong Seok Lee. "**Conserved PDE4D short-isoform bias defines pathogenic Th1.17 state across human inflammatory diseases.**" *Nature Immunology* 2026. in revision

I analyzed 251,055 memory CD4-positive T cells from four cohorts of 16 donors each, totaling 64 donors. Consensus NMF identified 12 programs. I wrote UMI-aware code to resolve PDE4D isoforms from 5' reads. The cell state recurred across 702 atlas samples, and held-out arthritis classification reached AUC 0.780. The study found that short-isoform overexpression increased the pathogenic fraction and selective inhibition reduced it.

- 3 **Seong-Jun Kang**, Yong-Hee Kim, Thuy Nguyen-Phuong, Yijoon Kim, Jin-Mi Oh, Jae-chun Go, DaeSik Kim, Chung-Gyu Park, Hyunsu Lee, Hyun Je Kim. "**Immune cell-enriched single-cell RNA sequencing unveils the interplay between infiltrated CD8⁺ T resident memory cells and choroid plexus epithelial cells in Alzheimer's disease.**" *Journal of Neuroimmunology* 2025. [Download PDF](#)

I carried out whole-brain processing, enzymatic digestion, Percoll enrichment, CD45-positive sorting, library preparation, and analysis in a 5xFAD mouse study. I analyzed 11,587 cells and identified ten immune populations. The study examined CD8-positive tissue-resident memory T cells and choroid plexus epithelium at the blood-CSF barrier.

- 4 Youngkyoung Lim, **Seong-Jun Kang**, Beom Keun Cho, Hyun Je Kim, ... Chong Hyun Won, Chung-Gyu Park. "Intra-tumoral heterogeneity and immune escape of melanoma arising from congenital melanocytic nevus revealed by spatial gene expression profiling." **JEADV** 2022. [Download PDF](#)

I ran GeoMx assay setup, instrument operation, immunofluorescence staining, region selection, barcode collection, and spatial analysis. One melanoma provided 16 whole-transcriptome regions of interest across four regions. The analysis measured 18,677 genes and showed lower HLA class I and II expression with greater invasion depth.

TECHNICAL EXPERTISE

Binding kinetics and affinity	Hands-on Biacore SPR for association and dissociation kinetics, affinity ranking, receptor blockade, and human/non-human-primate cross-reactivity. ELISA binding, competition, and epitope analysis. Antibody candidate comparison and lead selection.
Therapeutic lead development	Humanized anti-CD40 antibody program from candidate comparison through non-human-primate studies. Assay design, external study coordination, raw-data review, regulatory documentation, and candidate-advancement decisions.
External studies and collaboration	CRO and core-facility study design, assay specifications, data review, and milestone tracking. Participation in outsourced antibody and AAV manufacturing collaborations involving Samsung Biologics and WuXi. Participation in early joint-development research with Merck, CSL, and Johnson & Johnson.
Cell, tissue & immunological assays	Primary human T-cell isolation, expansion and functional assays; monocyte and dendritic-cell differentiation; mesenchymal stem cell culture. Enzymatic tissue digestion, Percoll enrichment, cell and nucleus isolation. High-dimensional flow cytometry (panels up to 24 markers); multiplex panel design; 10-color 6-way FACS. Olink targeted plasma proteomics (immune & neurology panels).
Single-cell / spatial / multi-omic	scRNA-seq, scTCR-seq, snRNA/snATAC; 10x 5'/3' library generation and 10x Chromium Connect automated preparation. Seurat , Scanpy , Squidpy , Monocle3 , CellChat/NicheNet ; scVI/scANVI, batch integration, consensus NMF ; Milo KNN-graph differential abundance, GLIPH, InferCNV. Custom UMI-aware, BAM-level isoform analysis code (Python). Spatial: GeoMx DSP run end to end, Visium , MACSima cyclic multiplexed immunofluorescence.
Computational	Python: scikit-learn, XGBoost/LightGBM/CatBoost, PyTorch ; AUC-validated classifiers held out on independent cohorts; multimodal deep-learning integration (imaging + transcriptomic + clinical). Linux servers · R/RStudio · Git-based reproducible analysis.
Microscopy & animal models	Spinning-disk confocal (Nikon CSU-W1 SORA), point-scanning confocal (Leica SP8), two-photon; IHC/IF (frozen & FFPE) with quantitative image analysis. Non-human primates across three species (rhesus, cynomolgus, marmoset), including EAE and islet-transplantation studies; mouse immune profiling.
Vision-language and retrieval models	Finetuning . LoRA adaptation of an open vision-language model (Gemma 4 12B , mlx-vlm) so that an amyloid pathology tile returns a structured JSON reading, with the schema enforced during decoding and the split held out at the case level. Against the same model zero-shot on 800 held-out tiles, cored-plaque F1 went from 0.12 to 0.44 and cerebral amyloid angiopathy from 0.00 to 0.69. Retrieval . A retrieval-augmented question-answering service written end to end: corpus chunking, BM25 and dense-vector retrieval fused by reciprocal rank, query expansion, and streamed answers that carry their sources. FastAPI , containerized, on Google Cloud Run ; it answers questions about this CV at kangseongjun.com.

PUBLICATIONS

18

17

1

PUBLISHED ARTICLES

- 2025 **Seong-Jun Kang**, Yong-Hee Kim, Thuy Nguyen-Phuong, Yijoon Kim, Jin-Mi Oh, Jae-chun Go, DaeSik Kim, Chung-Gyu Park, Hyunsu Lee, Hyun Je Kim. "Immune cell-enriched single-cell RNA sequencing unveils the interplay between infiltrated CD8⁺ T resident memory cells and choroid plexus epithelial cells in Alzheimer's disease." *Journal of Neuroimmunology*. [Download PDF](#)
- 2025 **Seong-Jun Kang**, Jeong-Ryeol Gong, Seon-Pil Jin, Jin-Mi Oh, Hyunjin Jin, Yuji Lee, Yewon Moon, Dongjun Kim, Hyo Jeong Nam, Hyun Seung Choi, Sanha Hwang, Yun Jung Huh, Kyung Yeon Han, Jihwan Moon, Jongsuk Chung, Woong-Yang Park, Chung-Gyu Park, Hyun Je Kim, Jeong Eun Kim. "Deciphering Dysfunctional Regulatory T Cells in Atopic Dermatitis." *Allergy*. [Download PDF](#)
- 2024 Jong-Min Kim, **Seong-Jun Kang**, So-Hee Hong, Hyunwoo Chung, Jun-Seop Shin, Byoung-Hoon Min, Hyun Je Kim, Jongwon Ha, Chung-Gyu Park. "Long-term control of diabetes by tofacitinib-based immunosuppressive regimen after allo-islet transplantation in diabetic rhesus monkeys that rejected previously transplanted porcine islets." *Xenotransplantation*. [Download PDF](#)
- 2022 Youngkyoung Lim, **Seong-Jun Kang**, Beom Keun Cho, Hyun Je Kim, ... Chong Hyun Won, Chung-Gyu Park. "Intra-tumoral heterogeneity and immune escape of melanoma arising from congenital melanocytic nevus revealed by spatial gene expression profiling." *JEADV*. [Download PDF](#)
- 2019 Jong-Min Kim, Jun-Seop Shin, Byoung-Hoon Min, **Seong-Jun Kang**, Il-Hee Yoon, Hyunwoo Chung, Jiyeon Kim, Eung-Soo Hwang, Jongwon Ha, Chung-Gyu Park. "JAK3 inhibitor-based immunosuppression in allogeneic islet transplantation in cynomolgus monkeys." *Islets*. [Download PDF](#)
- 2026 Jung Ho Lee, Sung Ha Lim, Hyungdon Kook, **Seong-Jun Kang**, Geon Lee, Brian Hyohyoung Lee, Hyunsung Nam, YongJun Kim, Christine Suh-Yun Joh, Soyoung Jeong, Dongryeol Shin, Hyun Je Kim, Jiyoung Ahn. "Mycosis Fungoides-like Atopic Dermatitis Represents a Th22-dominant Inflammatory Endotype." *Allergy*.
- 2024 Brian Hyohyoung Lee, Yoon Ji Bang, Sung Ha Lim, **Seong-Jun Kang**, Sung Hee Kim, Seunghee Kim-Schulze, Chung-Gyu Park, Hyun Je Kim, Tae-Gyun Kim. "High-dimensional profiling of regulatory T cells in psoriasis reveals an impaired skin-trafficking property." *eBioMedicine*. [Download PDF](#)
- 2024 Youngkyoung Lim, Beom Keun Cho, **Seong-Jun Kang**, Soyoung Jeong, Hyun Je Kim, Jiyeon Baek, Ji Hwan Moon, Cheol Lee, Chan-Sik Park, Je-Ho Mun, Chong Hyun Won, Chung-Gyu Park. "Spatial transcriptomic analysis of tumour-immune cell interactions in melanoma arising from congenital melanocytic nevus." *JEADV*. [Download PDF](#)
- 2023 Sunyoung Jung, Sunho Lee, Hyun Je Kim, Sueon Kim, Ji Hwan Moon, Hyunwoo Chung, **Seong-Jun Kang**, Chung-Gyu Park. "Mesenchymal stem cell-derived extracellular vesicles subvert Th17 cells by destabilizing ROR γ t through posttranslational modification." *Experimental & Molecular Medicine*.
- 2022 Sangho Lee, Han-Teo Lee, Young Ah Kim, Il-Hwan Lee, **Seong-Jun Kang**, Kyeongpyo Sim, Chung-Gyu Park, Kyungho Choi, Hong-Duk Youn. "The optimized core peptide derived from CABIN1 efficiently inhibits calcineurin-mediated T-cell activation." *Experimental & Molecular Medicine*.
- 2021 So Hee Hong, Hyun Je Kim, **Seong-Jun Kang**, Chung-Gyu Park. "Novel Immunomodulatory Approaches for Porcine Islet Xenotransplantation." *Current Diabetes Reports*. [Download PDF](#)

2020	Sunho Lee, Sueon Kim, Hyunwoo Chung, Ji Hwan Moon, Seong-Jun Kang , Chung-Gyu Park. "Mesenchymal stem cell-derived exosomes suppress proliferation of T cells by inducing cell cycle arrest through p27kip1/Cdk2 signaling." <i>Immunology Letters</i> .
2019	Hyun Je Kim, Ji Hwan Moon, Hyunwoo Chung, Jun-Seop Shin, Bongi Kim, Jong Min Kim, Jung-Sik Kim, Il-Hee Yoon, Byoung-Hoon Min, Seong-Jun Kang , Yong-Hee Kim, Kyuri Jo, Joungmin Choi, Heejoon Chae, Won Woo Lee, Sun Kim, Chung-Gyu Park. "Bioinformatic analysis of peripheral blood RNA-sequencing sensitively detects the cause of late graft loss following overt hyperglycemia in pig-to-non-human primate islet xenotransplantation." <i>Scientific Reports</i> .
2018	Byoung-Hoon Min, Jun-Seop Shin, Jong-Min Kim, Seong-Jun Kang , Hyun-Je Kim, Il-Hee Yoon, Su-Kyoung Park, Ji-won Choi, Min-Suk Lee, Chung-Gyu Park. "Delayed revascularization of islets after transplantation by IL-6 blockade in pig-to-non-human primate islet xenotransplantation model." <i>Xenotransplantation</i> .
2018	Jun-Seop Shin, Jong-Min Kim, Byoung-Hoon Min, Il Hee Yoon, Hyun Je Kim, Jung-Sik Kim, Yong-Hee Kim, Seong-Jun Kang , Jiyeon Kim, Hee-Jung Kang, Dong-Gyun Lim, Eung-Soo Hwang, Jongwon Ha, Sang-Joon Kim, Wan Beom Park, Chung-Gyu Park. "Pre-clinical results in pig-to-non-human primate islet xenotransplantation using anti-CD40 antibody (2C10R4)-based immunosuppression." <i>Xenotransplantation</i> .
2018	Jong-Min Kim, Jun-Seop Shin, Byoung-Hoon Min, Il Hee Yoon, Seong-Jun Kang , Won-Young Jeong, Sang-Joon Kim, Chung-Gyu Park. "Pre-Clinical Results of Islet Allo-Transplantation Using JAK Inhibitor as Replacement for Tacrolimus in Cynomolgus Monkeys." <i>Transplantation</i> .
2018	Jung-Sik Kim, Hyunwoo Chung, Nari Byun, Seong-Jun Kang , Sunho Lee, Jun-Seop Shin, Chung-Gyu Park. "Construction of EMSC-islet co-localizing composites for xenogeneic porcine islet transplantation." <i>Biochemical and Biophysical Research Communications</i> .

MANUSCRIPT IN REVISION

2026	Woogil Song, Seong-Jun Kang , Taehee Stella Kim, Seyoung Jung, Sung Hee Kim, Su Chan Park, Sung Hae Chang, Eun Young Lee, Tae-Gyun Kim, Hyun Je Kim, Jeong Seok Lee. "Conserved PDE4D short-isoform bias defines pathogenic Th1.17 state across human inflammatory diseases." <i>Nature Immunology</i> . In revision
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MANUSCRIPTS IN PREPARATION

2026	Kangsan Roh, Sabyasachi Das, Seong-Jun Kang , Olga Barreiro, Rodrigo Javier Gonzalez, Timothy P. Padera, Shraddha Shirguppe, Ahlim Lee, Youngju Song, Pazhanichamy Kalailingam, Claire E. Fong, Aarushi Gandhi, Meehyun Kim, Diana Tambala, Adam Vernon Crain, Casey A. Maguire, Ulrich H. Von Andrian, Christian L. Lino Cardenas, Sekeun Kim, Benjamin P. Kleinstiver, Mark E. Lindsay, Patricia L. Musolino, Rajeev Malhotra. "In-vivo base editing of ACTA2 R179H restores lymphatic function and immune cell homeostasis in a murine model of Multisystemic Smooth Muscle Dysfunction Syndrome (MSMDS)." In preparation
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BOOKS

강성준 (Seong-Jun Kang), 김도균, 한진선. **의료 바이오 연구자를 위한 AI 신약개발 실전 가이드**. Beommun Education (범문에듀케이션). Forthcoming September 21, 2026. 548 pages. ISBN 979-11-5943-583-6.

English title translation: *AI Drug Discovery: A Practical Guide for Medical and Biomedical Researchers*. Publisher

The book covers molecular and protein data preparation, QSAR, ChemBERTa, graph neural networks, docking, virtual screening, candidate optimization, and follow-up experimental validation.

■ SELECTED GRANT WRITING & PROGRAM FUNDING

INDUSTRY · PB IMMUNE THERAPEUTICS (MANAGER – ASSOCIATE DIRECTOR)

FUNDING AGENCY / SPONSOR	PROJECT	PERIOD	BUDGET (USD)	ROLE / CONTRIBUTION
Korea Drug Development Fund (KDDF)	Anti-CD40 antibody lead selection and optimization (multiple sclerosis)	2021–2023	~\$571,000	Lead proposal writer (>90% of proposal & execution)
Korea Drug Development Fund (KDDF)	Anti-CD40 lead compound (PB101): efficacy and pharmacodynamic biomarkers (pemphigus vulgaris)	2022–2024	~\$321,000	Lead proposal writer (>90% of proposal & execution)
Ministry of SMEs and Startups (MSS)	Scale-up Funding for Tech Commercialization	2022–2023	~\$222,000	Lead proposal writer; managed >90% of execution
Ministry of SMEs and Startups (MSS)	Early-stage startup R&D: anti-CD40 biologic (PB-40N) to prevent anti-drug antibodies	2021–2022	~\$74,000	Lead proposal writer & core researcher
NAVER Corporation	Engineered regulatory T cells for immune-mediated diseases (PB20X_Treg)	2023–2026	~\$128,000	Proposal & study design (~50%)

DOCTORAL TRAINING · SEOUL NATIONAL UNIVERSITY

FUNDING AGENCY / SPONSOR	PROJECT	PERIOD	BUDGET (USD)	ROLE / CONTRIBUTION
Ministry of Science and ICT (MSIT)	Multi-layered immune-disease single-cell map for molecular target discovery (SCAID, A9-22-1-05)	2022–2028 (3+3)	~\$1.6M	Participated in proposal writing (>30%)
Ministry of Education (MOE) & Ministry of Science and ICT (MSIT)	Laboratory-Specialized Tech-Transfer & R&BD Grant	2020–2021	~\$40,000	Grant writing, experimental design & technical validation

I participated in proposal writing and project execution for the awards listed above. Budgets are approximate conversions at KRW 1,400 per USD. Project periods are award periods, not appointment dates.

■ AWARDS, PATENTS & SCHOLARSHIPS

PATENTS

2024	Antagonistic Anti-CD40 Humanized Antibody. Republic of Korea, Appl. No. 10-2024-0085972; registered 2024-07-01. Co-inventor.
2019	Immunosuppression Composition Comprising JAK Inhibitor. Republic of Korea, Appl. No. 10-2014-3490000; registered 2019-08-20.

AWARDS

2024	Grand Prize , 2024 GPTers AI Hackathon (Project GAIA Team)
2023	Grand Prize , SBA Smart Workathon (Work Process Innovation)
2017	Best Poster Award , American Transplant Congress, Chicago · Best Presentation Award , Korean Association of Immunologists

SCHOLARSHIPS

2014–2020	NRF Research Scholarship for Ph.D. Candidates · Dean's Lecture & Research Support Scholarship, SNU College of Medicine · BK21 Plus Research Scholarship
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■ SELECTED PRESENTATIONS

Poster KAI Annual Meeting, Incheon (2022) · Cytokine, Hawaii (2022) · American Transplant Congress, Chicago (2017) · The Transplantation Society, Hong Kong (2016) · IPITA/IXA/CTS, Melbourne (2015) · TREG, Shanghai (2012)

■ REFERENCES

Chung-Gyu Park, M.D., Ph.D.

Professor, Microbiology & Immunology / Biomedical Sciences,
SNU College of Medicine; CEO/CSO, PB Immune Therapeutics
· chgpark@snu.ac.kr

*Doctoral advisor and industry supervisor; directed the anti-CD40
antibody program.*

Seung-Kuy Cha, Ph.D.

Professor, Dept. of Physiology & Global Medical Science,
Yonsei University Wonju College of Medicine ·
skcha@yonsei.ac.kr

Most recent research supervisor (2025–2026).

Hyun Je Kim, M.D., Ph.D.

Assistant Professor, Microbiology & Immunology / Biomedical
Sciences; Interdisciplinary Program in Artificial Intelligence
(IPAI), SNU · tte9801@snu.ac.kr

*Co-corresponding on the single-cell studies; supervised the
computational work.*

Jeong Seok Lee, M.D., Ph.D.

Associate Professor, Graduate School of Medical Science &
Engineering, KAIST, Daejeon · chemami@kaist.ac.kr

Corresponding author, PDE4D / Nature Immunology study.

Seong-Jun Kang · Curriculum Vitae · Keating Lab Available from August 2026 · based in Boston, MA · currently authorized to work in the United States without employer sponsorship to begin employment Updated September 2026